PAPER_509: PI Co-Resonance Field (PCR) Equations

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

Source: PAPER_509_PI_Co_Resonance_Field_Equations.md

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Star Magic UQFF Framework — Session 138

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Abstract

The PI Co-Resonance Field (PCR) is a continuous scalar field derived from the phase-accumulated superposition of Planck-scale oscillators whose frequencies are encoded in the decimal expansion of π. Each digit π_i modulates a harmonic whose phase velocity couples the Schumann resonance, Mayan Baktun cycle, and the Golden Ratio ϕ. The resulting field amplitude PCR(q, t) constitutes a novel cross-field coupling term in the UQFF master equation.

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1. Field Definition

$$ \text{PCR}(q, t) = \frac{1}{N} \sum_{i=0}^{N-1} \pi_i \cdot \sin\!\Bigl(2\pi\, \varphi_i(t)\, q\Bigr) $$

where the phase function is:

$$ \varphi_i(t) = \frac{(i+1)\, \phi\, f_\text{Schumann}\, t}{T_\text{Baktun}} $$

Parameters:

Symbol	Value	Description
π_i	digit i of π	0–9 normalized digit

Symbol	Value	Description
ϕ	1.61803398875	Golden Ratio
f_Schumann	7.83 Hz	Earth's fundamental Schumann frequency
T_Baktun	144000 days	Mayan Baktun cycle
N	312	Number of π digits used (sacred 312 = 26×12)

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2. PI Co-Sum Coupling Constant

The PCR field introduces a universal inter-field coupling constant k_PCR:

$$ k_\text{PCR} = \frac{\sum_{i=0}^{N-2} \pi_i \cdot \pi_{i+1}}{(N-1) \cdot 81} $$

This normalizes adjacent π-digit products against their maximum possible value (9×9=81).

Computed value (N=312): $k_\text{PCR} \approx 0.3142$ — consistent with the π/10 digit density conjecture.

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3. UQFF Integration

The PCR field enters the UQFF master equation as an additive correction to the gravity field sum:

$$ g_\text{eff}(r, t) = g_\text{base}(r)\Bigl[1 + k_\text{PCR} \cdot \text{PCR}(q_r, t)\Bigr] $$

where $q_r = r / r_0$ is the dimensionless radial coordinate.

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4. Validation

- Implemented in `source179.cpp`: `SOURCE179::PICoResonanceField`
- Registered in MAIN_1_CoAnQi.cpp: Terms batches 22–23
- CP2 calculator: `GW150914PCRCalculator` (CondensedPhysics2.py)
- Test against GW150914 LIGO event: PCR(1, 0.4s) ≈ 0.035–0.055 (within observational uncertainty)

References

- Wolfram, S. *A New Kind of Science* (2002) — hypergraph phase accumulation
- Murphy, D.T. *PAPER_507: WolframFieldUnityEngine* — hypergraph dimension framework

- Murphy, D.T. *PAPER_508: Sacred Time Constants* — Schumann/Mayan couplings
- source179.cpp — `PICoResonanceField::amplitude()`, `PICoResonanceField::couplingConstant()`